

TweetBot using OpenWhisk

In order to use TweetBot in a serverless approach, you need to install OpenWhisk in your machine and have a Bluemix account. For installing OpenWhisk, refer to <https://console.bluemix.net/openwhisk/cli> for the set-up and configuration information. The beauty of serverless technology is that you don't have to rewrite your entire application; just tweak the plain code that runs in your machine and you'll be fine! Surprisingly, our *TweetBot* requires only one change — it should have a main function with an input parameter (dictionary type) and is to be saved as a `__main__.py` file.

Here is the modified Python code:

```
# Program to retweet five latest tweets based on a hashtag
(#Serverless)
from TwitterFollowBot import TwitterBot

def retweet(dict):
    # create an instance of the TwitterFollowBot
    # by default, the bot will look for a configuration file
    # called config.txt
    # in your current directory
    my_bot = TwitterBot()
    # autoretweets the 5(count)
    # latest tweets that matches the hashtag
    my_bot.auto_tweet("#Serverless", count = 5)

def main(dict):
    retweet(dict)
    return {'message': 'retweeted successfully'}
```

Now, save this Python code in a file named `__main__.py`. Create the `config.txt`, `already-followed.txt`, `followers.txt` and `following.txt` (as earlier), and zip them all with the `__main__.py` file and the `TwitterFollowBot` dependency module files.

Invocation process

Once the `wsk` CLI (command line interface) is installed and the zip file is ready, follow the steps given below.

Step 1. Create and update an action: Log in to the IBM Bluemix account (<https://console.bluemix.net/openwhisk/>) and create a new action. You can upload the zip files with dependencies only via the CLI. The syntax is:

```
wsk action create <action-name> --kind <language:version>
<file_name>
Sample Command:
wsk action create tweetBot --kind python:3 OpenWhisk.zip
```

Step 2. Invoke the function (non-blocking mode):
The syntax is:

```
wsk action invoke <action-name>
Sample Command: wsk action invoke tweetBot
```



Figure 4: Invocation process

```
PS D:\> wsk action create tweetBot --kind python:3 openwhisk.zip
PS D:\> wsk action invoke tweetBot
PS D:\> wsk action invoke tweetBot
PS D:\> wsk action invoke tweetBot
PS D:\> wsk activation result f4df0d1dcb12488396978d2cda832b09
{
  "message": "retweeted successfully"
}
PS D:\>
```

Figure 5: Result of the command

Step 3. Check for the result: Since we invoked the function in a non-blocking mode (because we haven't added the '—blocking' parameter), the command returned immediately, but it is executing in the background. The syntax is:

```
wsk activation result <action ID>
Sample Command:
wsk activation result f4df0d1dcb12488396978d2cda832b09
```

Step 4. Check out the logs: The syntax is:

```
wsk activation logs <action ID>
Sample Command:
wsk activation logs f4df0d1dcb12488396978d2cda832b09
```

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These are the best of times, and these are the worst of times! There are so many awesome new technologies to catch up on. But, we simply can't. We have seen a progression of computing models — from virtualisation, IaaS, PaaS, containers, and now, serverless — all in a matter of a few years. You certainly don't want to be left behind. So join us at the Serverless Summit, India's first confluence on serverless technologies, being held on October 27, 2017 at Bengaluru. It is the best place to hear from industry experts, network with technology enthusiasts, as well as learn about how to adopt serverless architecture. The keynote speaker is John Willis, director of ecosystem development at Docker and a DevOps guru (widely known for the book 'The DevOps Handbook' that he co-authored). *Open Source For You* is the media partner and the Cloud Native Computing Foundation is the community partner for this summit. For more details, please visit the website www.inserverless.com.

Automate the invocation

The moment the action was invoked, your Twitter account has been retweeted the five latest tweets that have the hashtag '#Serverless' in it. However, this is still a manual invocation. For maximum impact, it would be better to